

LARGE ERDŐS–STRAUS WITNESSES II: THE MULTIPLIER MODULUS IS INFINITELY OFTEN $\exp((\log p)^c)$

ANONYMOUS

ABSTRACT. For a prime p let $W(p)$ be the least modulus $M \equiv 3 \pmod{4}$ of a “multiplier” congruence that yields a solution of $4/p = 1/x + 1/y + 1/z$. In earlier work we showed that $W(p)$ exceeds every fixed power of $\log p$ for infinitely many primes p in Mordell’s hard classes. Here we prove $W(p) \geq \exp(c(\log p)^{1/14})$ for infinitely many hard primes, assuming Elsholtz and Tao’s average bound for $\tau(kab^2 + 1)$; equivalently, the least hard prime with $W(p) > T$ is at most $\exp(O((\log T)^{14}))$. Without the Elsholtz–Tao input we get $\log W(p) \geq (1/(2 \log 2) - o(1)) \log \log p \cdot \log \log \log p$ infinitely often. As before, the class of one modulo the small primes leaves a residual congruence system on the larger primes. The local lemma shows that its avoiding set has Haar density $e^{-O(S)}$, where S is its total mass. To transfer this density to primes we need a *pointwise* minorant that is a short combination of congruence classes, each on few primes. We take the one-sided ℓ^2 sandwich of Bazzi and Razborov from the theory of bounded independence. Its error is the Efron–Stein tail of the avoidance indicator. After a binary encoding of residues the indicator is a CNF of small width, so Håstad’s switching lemma, through the argument of Linial, Mansour and Nisan, bounds the tail. The minorant is transferred to primes by the Linnik-range prime number theorem of Thorner and Zaman. No codegree condition, level structure or truncated inclusion–exclusion is needed.

1. INTRODUCTION

The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ is solvable in positive integers for every $n \geq 2$. It suffices to treat primes. It has been verified for $n \leq 10^{18}$ ([20]; [25] for 10^{17}), and Mordell’s identities settle every prime outside the six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$ [19] (cf. [22]). We call the primes in these classes *hard*. For the exceptional set see Vaughan [27] and [23]; for the average number of solutions see [9].

For $M \equiv 3 \pmod{4}$ put $A_M = (M + 1)/4$. If $uvw = A_M$ and $n \equiv -uv^{-1} \pmod{M}$, then $s = (nv + u)/M$ is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsuv} + \frac{1}{nuvw} \tag{1}$$

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(direct verification; this is the classical multiplier family, cf. [27, 9]). Define

$$W(n) = \min\{M \equiv 3 \pmod{4} : n \equiv -uv^{-1} \pmod{M} \text{ for some } uvw = A_M\}, \quad (2)$$

with $W(n) = +\infty$ if no such M exists. Thus $W(p) \leq T$ provides an explicit representation with modulus at most T . Every prime $p \equiv 1 \pmod{\text{lcm}\{M \leq T\}}$ has $W(p) > T$ (Lemma 2.2), so Linnik's theorem gives $W(p) \gg \log p$ infinitely often. In [2] we proved that for every fixed k , $W(p) \geq (\log p)^k e^{-C_k \log \log p / \log \log \log p}$ for infinitely many hard primes. Here we prove a sub-exponential rate.

Theorem 1.1 (Proved modulo Theorem 3.1 and [9, Prop. 1.4]). *There is an absolute constant $c > 0$ such that for infinitely many hard primes p*

$$W(p) \geq \exp(c(\log p)^{1/14}).$$

More precisely, for every sufficiently large T there is a prime $p \equiv 1 \pmod{840}$ with $p > T$, $W(p) > T$ and $\log p \leq C(\log T)^{14}$.

Theorem 1.2 (Proved modulo Theorem 3.1). *For infinitely many hard primes p ,*

$$\log W(p) \geq \left(\frac{1}{2 \log 2} - o(1) \right) \log \log p \cdot \log \log \log p.$$

Theorem 3.1 is Thorner and Zaman's prime number theorem in the Linnik range [26]. No hypothesis on Siegel zeros is needed. [9, Prop. 1.4] is used only to bound the total mass of the residual system by $(\log T)^4 \log \log T$ (Lemma 2.4). Without it, the same argument gives Theorem 1.2. All constants are effective, except that we have not checked whether the implied constant in [9, Prop. 1.4] is. We make no numerical claim.

The method. We fix T and impose $n \equiv 1$ modulo the prime powers $\leq T$ of a set Π of primes, consisting of the primes up to $z = (\log T)^2$ and a few further ones. Then $W(n) > T$ whenever n avoids a finite family of congruence *events*, each on at most $k \approx \log T / (2 \log \log T)$ of the remaining (*free*) primes (§2). In the Haar measure on residues the local lemma shows that the avoiding set has density $\delta \geq e^{-2.2S}$, where $S \ll (\log T)^4 \log \log T$ is the total mass, once an iterated quarantine has made the mass at each free prime small. To find a *prime* in the avoiding set we need a function $B = \sum_i c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ with $B \leq \mathbf{1}[\text{avoid}]$ pointwise, mean $\gg \delta$, controlled ℓ^1 mass, and moduli d_i that each involve few primes. The transfer theorem (Theorem 4.1, from [2]) then gives a prime p with $\log p \ll K \max(\log Z, K)$, where $K = 1 + \log(\ell^1/\text{mean})$ and Z is the largest modulus. Both are polynomial in $\log T$ in our construction.

In [2] the minorant was an alternating expansion (inclusion–exclusion truncated by the number of primes in the support), organised in levels. That route loses a factorial in the number of levels, which limits it to fixed powers of $\log p$. Here we use instead the one-sided ℓ^2 -approximation scheme of Bazzi

[3], in the form of Razborov [24] (Lemma 5.3). For events $E_1, \dots, E_m$ with indicators A_i and arbitrary functions u_j ,

$$B = 1 - \sum_i A_i \left(1 - \sum_{j < i} A_j u_j \right)^2 \leq \mathbf{1}[\text{no } E_i \text{ occurs}],$$

and $\mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$, where $F^{(j)}$ is the indicator that no earlier event occurs, conditioned on E_j . Choosing u_j as the best ℓ^2 approximation of $F^{(j)}$ by sums of functions of at most t coordinates makes every cell of B involve at most $3k + 2t$ primes, and makes the error an *Efron–Stein tail* of $F^{(j)}$. We bound this tail in §6. Encode each residue in binary. Then $1 - F^{(j)}$ becomes a DNF of width $k \log_2(4T^2)$, and Håstad’s switching lemma with the Linial–Mansour–Nisan argument [12, 17] gives tail $\leq 4 \cdot 2^{-k_0}$ at level $t \asymp k \log T \cdot k_0$ (Lemma 6.2). We need $k_0 \asymp S + k \log T$, so $t \ll (\log T)^6$ and $\log p \ll (\log T)^{14}$.

Relation to the literature. Lemma 5.3 is Razborov’s identity, and Lemma 6.2 is a routine application of [12, 17] (we follow [21, §4]). We claim no novelty for either. To our knowledge these tools from the theory of bounded independence (“polylogarithmic independence fools DNF and AC^0 ” [3, 24, 6]) have not been used before to construct sieve minorants for primes. The point is that, in the Linnik range, primes $p \equiv 1 \pmod{Q}$ distribute over congruence cells with a *relative* error. So the number of primes per cell is what costs, while the size of the moduli is nearly free. Earlier uses of the switching lemma in number theory known to us run in the opposite direction: they show that an arithmetic function such as the Möbius function is uncorrelated with bounded-depth circuits [10, 5]. Our literature search was limited, and we make no priority claim. The Haar-side input, a local-lemma lower bound for the density of the uncovered set of a congruence system, has well-known relatives in the work on Erdős covering systems by Hough [13] and by Balister, Bollobás, Morris, Sahasrabudhe and Tiba [4]. We do not use those results.

Status conventions. Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Cited* marks external statements; *Classical* marks standard results for which we include a proof or a precise textbook reference. Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large $W(p)$ may well have other representations.

Notation. T is large, $\mathcal{L} = \log T$, $\log_2$ denotes the binary logarithm only where explicitly said (otherwise $\log \log$ is written out), τ is the divisor function and $\tau^*(x) = \max_{n \leq x} \tau(n)$.

2. ATOMS AND THE RESIDUAL SYSTEM

For $M \equiv 3 \pmod{4}$ put $\mathcal{R}(M) = \{-4D \bmod M : D \mid A_M^2\}$. Note that $\gcd(A_M, M) = 1$ (as $4A_M = M + 1$), so every element of $\mathcal{R}(M)$ is a unit class.

Lemma 2.1 (Proved; cf. [2]). $\{-uv^{-1} \bmod M : uvw = A_M\} = \mathcal{R}(M)$. Consequently, for every integer n , $W(n) > T$ if and only if $n \bmod M \notin \mathcal{R}(M)$ for every $M \leq T$ with $M \equiv 3 \pmod{4}$.

Proof. Since $4A_M \equiv 1 \pmod{M}$, from $uvw = A_M$ we get $v^{-1} \equiv 4uw$, hence $-uv^{-1} \equiv -4u^2w$ and $D = u^2w$ divides A_M^2 . Conversely let $D \mid A_M^2$. For each prime q with $q^a \parallel A_M$ and $q^d \parallel D$ ($d \leq 2a$) put $u_q = \max(0, d-a)$, $w_q = d - 2u_q$, $v_q = a - u_q - w_q$; these are non-negative, so $u = \prod q^{u_q}$, $v = \prod q^{v_q}$, $w = \prod q^{w_q}$ satisfy $uvw = A_M$ and $u^2w = D$. $\square$

Lemma 2.2 (Class of one; proved). $1 \notin \mathcal{R}(M)$ for every $M \equiv 3 \pmod{4}$.

Proof. If $-4D \equiv 1$ then $D \equiv -A_M \pmod{M}$ (multiply by A_M). Since $4A_M^2 \equiv A_M$, the map $D \mapsto A_M^2/D$ preserves this congruence, so we may assume $D \leq A_M$; then $0 < D + A_M \leq 2A_M < M$, while $M \mid D + A_M$. $\square$

An *atom* is a pair (M, D) with $M \leq T$, $M \equiv 3 \pmod{4}$ and $D \mid A_M^2$. For a set Π of primes write $M = m_\Pi r_\Pi$, where m_Π is the largest divisor of M composed of primes in Π . The atom *survives* Π if $m_\Pi \mid 4D + 1$ and $r_\Pi > 1$; it then defines the *event* $n \equiv -4D \pmod{r_\Pi}$, a unit class, of Haar weight $1/\varphi(r_\Pi)$. Put

$$S^* = S^*(T) = \sum_{(M,D)} \max_{\Pi} \frac{\mathbf{1}[(M,D) \text{ survives } \Pi]}{\varphi(r_\Pi)},$$

the sum over all atoms and the maximum over all sets of primes.

Theorem 2.3 (Elsholtz–Tao [9, Prop. 1.4]; cited). For any $A, B > 1$ and any positive integer $\kappa \ll (AB)^{O(1)}$,

$$\sum_{a \leq A} \sum_{b \leq B} \tau(\kappa ab^2 + 1) \ll AB \log(A+B) \log(1+\kappa).$$

We use it only with $\kappa = 4$, so the factor $\log(1+\kappa) = \log 5$ is a constant.

Lemma 2.4 (Mass bound; proved, and (b) modulo [9, Prop. 1.4]). (a) $S^* \leq \exp((\log 2 + o(1))\mathcal{L}/\log \mathcal{L})$. (b) $S^* \ll \mathcal{L}^4 \log \mathcal{L}$.

Proof. Let (M, D) survive Π , put $A = A_M$ and $g = \gcd(M, 4D + 1)$. Then $m_\Pi \mid g$, and $\varphi(M) \leq m_\Pi \varphi(r_\Pi)$, so $1/\varphi(r_\Pi) \leq g/\varphi(M) \leq C g \log \mathcal{L}/M$ for an absolute C . This bound does not depend on Π , so $S^* \leq C \log \mathcal{L} \sum_{(M,D)} g/M$.

The map $D \mapsto D' = A^2/D$ preserves g . Indeed, D is a unit modulo M , and for a prime power $q^a \mid M$ with $q^a \mid 4D + 1$ we have $4D' \equiv 4A^2 \cdot (-4) \equiv -(4A)^2 \equiv -1 \pmod{q^a}$. It exchanges $D < A$ with $D > A$, so it suffices to sum over $D \leq A$ and double. Write $D = sr^2$ with s squarefree. Then $D \mid A^2$

if and only if $sr \mid A$; write $A = srh$, so that $D \leq A$ if and only if $r \leq h$, and $M = 4srh - 1 \geq 3srh$. From $g \mid 4sr^2 + 1$ and $g \mid M = 4srh - 1$ we get $g \mid 4sr(r + h)$, and $\gcd(g, 4sr) = 1$, so $g \mid \gcd(4sr^2 + 1, r + h)$. For fixed (s, r) the triple (M, D, g) is determined by h . Using $g \leq \sum_{e \mid g} e$, we get

$$\sum_{r \leq h \leq X} \frac{g}{h} \leq \sum_{e \mid 4sr^2 + 1} e \sum_{\substack{r \leq h \leq X \\ h \equiv -r \pmod{e}}} \frac{1}{h} \leq (3 + \log X) \tau(4sr^2 + 1), \quad X = \frac{T+1}{4}.$$

For the last step: the least admissible h is at least $\max(r, e - r) \geq e/2$, so the inner sum is at most $2/e + (1 + \log X)/e$. Hence

$$S^* \leq C' \log \mathcal{L} \cdot \mathcal{L} \sum_{sr^2 \leq X} \frac{\tau(4sr^2 + 1)}{sr}.$$

(a) Bound $\tau(4sr^2 + 1) \leq \tau^*(T + 2)$ and $\sum_{s, r \leq X} 1/(sr) \leq (1 + \mathcal{L})^2$. Then use $\log \tau^*(x) \leq (\log 2 + o(1)) \log x / \log \log x$ [14, Thm. 317]. (b) Split $s \in [2^i, 2^{i+1})$, $r \in [2^j, 2^{j+1})$ with $2^i, 2^j \leq X$. By Theorem 2.3 with $\kappa = 4$, $A = 2^{i+1}$, $B = 2^{j+1}$ (so $A, B > 1$ and $\kappa \ll (AB)^{O(1)}$), each block contributes $\ll 2^{-i-j} AB \log(A + B) \log 5 \ll \mathcal{L}$. There are $\ll \mathcal{L}^2$ blocks, so the sum is $\ll \mathcal{L}^3$. $\square$

Lemma 2.5 (Iterated quarantine; proved, cf. [2]). *Let $2 \leq z \leq T$, $k = \lfloor \mathcal{L} / \log z \rfloor$ and $0 < c_0 \leq 1$. For a set $\Pi \supseteq \{\ell \leq z\}$ and a prime $\ell \notin \Pi$ let $w_\ell(\Pi)$ be the total Haar weight of the distinct events of atoms surviving Π whose modulus r_Π is divisible by ℓ . Start from $\Pi_0 = \{\ell \leq z\}$ and repeatedly add to Π every prime $\ell \notin \Pi$ with $w_\ell(\Pi) > c_0$. The process stops at some $\Pi = \Pi_0 \cup \mathcal{B}$ with $|\mathcal{B}| \leq kS^*/c_0$, and then $w_\ell(\Pi) \leq c_0$ for every prime $\ell \notin \Pi$.*

Proof. The process stops since only primes $\leq T$ can have $w_\ell > 0$. If ℓ is added at a stage Π_i , then $c_0 < w_\ell(\Pi_i)$, which is at most the sum of the contributions to S^* of the atoms (M, D) with $\ell \mid M$. Summing over the added primes, each atom is counted at most once for each prime factor $\ell > z$ of M . There are at most k such factors since $M \leq T$. $\square$

Definition 2.6 (Residual system). Let $\Pi \supseteq \{\ell \leq z\}$ be as in Lemma 2.5, let $e_\ell \geq 1$ be maximal with $\ell^{e_\ell} \leq T$, and put $Q_\Pi = \text{lcm}(24, \ell^{e_\ell} : \ell \in \Pi, \ell \leq T)$. The *free* primes are $\mathcal{P} = \{\ell \leq T : \ell \notin \Pi\}$, with *coordinates* $X_\ell = n \bmod \ell^{e_\ell}$ taking values in $G_\ell = (\mathbb{Z}/\ell^{e_\ell})^\times$. The *Haar measure* $\mathbb{P}$ makes the X_ℓ independent and uniform. For each *distinct* pair $(r, c) = (r_\Pi, -4D \bmod r_\Pi)$ arising from an atom (M, D) surviving Π , with $r = \prod \ell^{a_\ell}$, and each *lift*, i.e. each $\sigma \in \prod_{\ell \mid r} G_\ell$ with $\sigma_\ell \equiv c \pmod{\ell^{a_\ell}}$, we take the event $E = \{X_\ell = \sigma_\ell \text{ for all } \ell \mid r_\Pi\}$, with support $\text{supp } E = \{\ell \mid r\}$. Let $E_1, \dots, E_m$ list these events (two pairs may have a common lift; such duplicates are kept), and let $F = \mathbf{1}[\text{no } E_i \text{ occurs}]$.

The lifts of one pair (r, c) partition the class $c \bmod r$, so their weights add up to $1/\varphi(r)$.

Lemma 2.7 (Properties of the residual system; proved). *In Definition 2.6, with $z \geq 7$:*

- (i) $840 \mid Q_\Pi$ and $\log Q_\Pi \leq (\pi(z) + |\mathcal{B}|)\mathcal{L} + \log 24$;
- (ii) $|\mathcal{P}| \leq T$, $|G_\ell| \leq T$, $|\text{supp } E_i| \leq k$, and $m \leq T^{k+2}$;
- (iii) $w_\ell := \sum_{i: \ell \in \text{supp } E_i} \mathbb{P}(E_i) = w_\ell(\Pi)$ for every $\ell \in \mathcal{P}$, and $S := \sum_{i \leq m} \mathbb{P}(E_i) \leq S^*$; so if Π is the output of Lemma 2.5, then $w_\ell \leq c_0$ for all $\ell \in \mathcal{P}$;
- (iv) (Reduction.) If $n \equiv 1 \pmod{Q_\Pi}$ and $F(n) = 1$ (the residues $X_\ell = n \bmod \ell^{e_\ell}$ need not be units), then $W(n) > T$.

Proof. (i) $24 \mid Q_\Pi$ and $5, 7 \in \Pi$. (ii) An event's support consists of the prime factors $> z$ of some $M \leq T$, so there are at most k of them. There are at most T^2 atoms, since $\tau(A_M^2) \leq 2A_M \leq T$, and each has at most $\prod_{\ell \mid r_\Pi} \ell^{e_\ell - a_\ell} \leq T^k$ lifts. (iii) Lifting preserves weights. Each pair (r, c) comes from an atom whose contribution to S^* is at least $1/\varphi(r)$. (iv) Let $M \leq T$ with $M \equiv 3 \pmod{4}$, and suppose $n \equiv -4D \pmod{M}$ with $D \mid A_M^2$. Every $\ell^v \parallel M$ with $\ell \in \Pi$ has $\ell^v \leq T$, so $m_\Pi \mid Q_\Pi$ and $-4D \equiv n \equiv 1 \pmod{m_\Pi}$, i.e. $m_\Pi \mid 4D + 1$. If $r_\Pi = 1$ this contradicts Lemma 2.2. Otherwise the atom survives Π , and $n \bmod \ell^{e_\ell}$ (a unit, as $-4D$ is) is a lift of $-4D \bmod \ell^{a_\ell}$ for each $\ell \mid r_\Pi$. So a lift of the pair $(r_\Pi, -4D \bmod r_\Pi)$ occurs, contradicting $F(n) = 1$. Conclude with Lemma 2.1. $\square$

3. THE ANALYTIC INPUT AND THE TRANSFER THEOREM

Theorem 3.1 (Thorner–Zaman [26, Cor. 1.4 and Rem. 1.5]; cited). *There are absolute effectively computable constants $c_4 > 0$, C_0 such that for coprime a, q with $q \geq 2$ and $x \geq q^{12}$,*

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod{q}}} \log p = \frac{\lambda x}{\varphi(q)} (1 + \varepsilon),$$

$$|\varepsilon| \leq C_0 \left(e^{-c_4 \log x / \log q} + e^{-c_4 (\log x)^{3/5} / (\log \log x)^{1/5}} \right),$$

where $\lambda = 1 - \chi_1(a)x^{\beta_1-1}/\beta_1$ if the exceptional zero β_1 for the modulus q exists and $\lambda = 1$ otherwise. Here, following McCurley [18] as quoted in [26, p. 1], $\prod_{\chi \bmod q} L(s, \chi)$ has no zero in $\Re s \geq 1 - 1/(13 \log(q(|\Im s| + 3)))$ apart from at most one real zero β_1 , which is simple and belongs to a unique real character χ_1 .

In Theorem 3.1 one has $0 < \lambda < 2$ (this is not part of the cited statement). Indeed $\beta_1 \geq 1 - 1/(13 \log 6) > 1/2$ and $\log x \geq 12 \log 2 > 2$, so $\log \beta_1 \geq 1 - 1/\beta_1 > 2(\beta_1 - 1) > (\beta_1 - 1) \log x$, i.e. $x^{\beta_1-1}/\beta_1 < 1$; and $|\chi_1(a)| \leq 1$.

We used the arXiv version; we did not compare the numbering with the published version. We also use the quoted McCurley statement for moduli other than those of Theorem 3.1; alternatively the classical Landau–Page theorem [7, Ch. 14] with a constant $c \leq 1/13$ in place of $1/13$ works verbatim.

Lemma 3.2 (One severe character per family; proved modulo the McCurley statement). *Let $\mathcal{Q}$ be a finite set of moduli, all $\leq Z$. Call (χ^*, β) severe if χ^* is a real primitive character whose conductor q^* divides some $q \in \mathcal{Q}$, $L(\beta, \chi^*) = 0$ and $\beta \geq 1 - 1/(13 \log(3Z^2))$. Then there is at most one severe pair. If (χ^*, β^*) is severe and $q^* \mid q \in \mathcal{Q}$, then $\beta_1 = \beta^*$ and χ_1 is induced by χ^* for the modulus q . If $q \in \mathcal{Q}$ and no severe pair has $q^* \mid q$, then any exceptional β_1 of q satisfies $x^{\beta_1-1}/\beta_1 \leq 2 \exp(-\log x/(13 \log 3Z^2))$.*

Proof. If $(\chi_a, \beta_a), (\chi_b, \beta_b)$ are severe with conductors dividing $q_a, q_b \in \mathcal{Q}$, put $q' = \text{lcm}(q_a, q_b) \leq Z^2$. Both characters induce characters mod q' with the same zeros in $\Re s > 0$, and both β lie in McCurley's region for q' ; uniqueness there gives $\beta_a = \beta_b$ and $\chi_a = \chi_b$. The second claim holds because β^* lies in McCurley's region for $q \leq Z$. For the third, an exceptional zero of q in $[1 - 1/(13 \log 3Z^2), 1)$ would be severe; also $\beta_1 \geq 1 - 1/(13 \log 6) > 1/2$. $\square$

4. A TRANSFER THEOREM

Theorem 4.1 (Proved modulo Theorem 3.1; from [2]). *Let $T \geq 1$, $Q \geq 2$, and let $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ be a finite real combination with $\gcd(d_i, Q) = \gcd(b_i, d_i) = 1$ and $B(n) \leq \mathbf{1}[W(n) > T]$ for all $n \equiv 1 \pmod{Q}$. Put $Z = Q \max d_i$, $\mu = \sum c_i/\varphi(d_i)$, $M_1 = \sum |c_i|/\varphi(d_i)$, and assume $\mu > 0$ and the twist condition: for every real primitive character ψ of conductor $f > 1$ with $\gcd(f, Q) = 1$ and $f \mid d_i$ for some i ,*

$$|\mu_\psi| \leq \mu/4, \quad \mu_\psi = \sum_{i: f \mid d_i} c_i \psi(b_i)/\varphi(d_i).$$

Let $K = 1 + \log(M_1/\mu)$. There is an absolute C_1 such that if $\log x \geq C_1 K \max(\log Z, K)$, then some prime $p \leq x$, $p \equiv 1 \pmod{Q}$, has $W(p) > T$.

Proof. Let $q_i = Qd_i$ and let a_i be the class $\equiv 1 \pmod{Q}$, $\equiv b_i \pmod{d_i}$; it is a unit class. Then $\sum_{p \leq x, p \equiv 1 \pmod{Q}} \log p B(p) = \sum_i c_i \theta(x; q_i, a_i)$. Since $M_1 \geq \mu$ we have $K \geq 1$, so taking $C_1 \geq 12$ gives $\log x \geq 12 \log Z$, i.e. $x \geq Z^{12} \geq q_i^{12}$, and Theorem 3.1 applies to every term. Put $E_1 = e^{-c_4 \log x / \log Z} + e^{-c_4 (\log x)^{3/5} / (\log \log x)^{1/5}}$ and $E_2 = e^{-\log x / (13 \log 3Z^2)}$. Since $\gcd(Q, d_i) = 1$ we have $\varphi(q_i) = \varphi(Q)\varphi(d_i)$, and $q_i \leq Z$, so

$$\sum_i c_i \theta(x; q_i, a_i) = \frac{x}{\varphi(Q)} \sum_i \frac{c_i \lambda_i (1 + \varepsilon_i)}{\varphi(d_i)}, \quad |\varepsilon_i| \leq C_0 E_1,$$

with λ_i the factor of Theorem 3.1 for (q_i, a_i) . Apply Lemma 3.2 to $\mathcal{Q} = \{q_i\}$; a severe β satisfies $\beta \geq 1 - 1/(13 \log 6) > 1/2$. Write the conductor of a severe χ^* as $q^* = q_Q^* q^{*'} with q_Q^* supported on the primes of Q and $\gcd(q^{*'}, Q) = 1$, and $\chi^* = \chi_Q^* \chi^{*'}$.$

Case A: a severe pair exists with $q^ \mid Q$.* Then $q^* \mid q_i$ for every i . By the second claim of Lemma 3.2, every λ_i is formed with the same pair (χ^*, β^*) , so $\lambda_i = 1 - \chi^*(a_i) x^{\beta^*-1}/\beta^* = 1 - x^{\beta^*-1}/\beta^* =: \lambda_*$, because $a_i \equiv 1 \pmod{q^*}$. Moreover $\lambda_* > 0$: since $\log x \geq 12 \log Z > 2$ and $1/2 < \beta^* < 1$,

$\log \beta^* \geq 1 - 1/\beta^* > 2(\beta^* - 1) > (\beta^* - 1) \log x$, i.e. $x^{\beta^*-1}/\beta^* < 1$. So $\sum c_i \theta \geq (x\lambda_*/\varphi(Q))(\mu - C_0 E_1 M_1)$.

Case B: otherwise. If a severe χ^* exists, then q^* divides some Qd_j with $\gcd(d_j, Q) = 1$, so $q^* \mid Q$. Hence $q^* \mid q_i$ iff $q^{*'} \mid d_i$, and in Case B we have $q^{*'} > 1$. For such i , $\chi_1 = \chi^*$ on units mod q_i (Lemma 3.2), and, as $a_i \equiv 1 \pmod{q_Q^*}$ and $a_i \equiv b_i \pmod{q^{*'}}$, $\chi^*(a_i) = \chi_Q^*(1)\chi^{*'}(b_i) = \psi(b_i)$ with $\psi = \chi^{*'}$. So $\lambda_i = 1 - \psi(b_i)x^{\beta^*-1}/\beta^*$. This ψ is real primitive (the coprime-conductor factors of a real primitive character are real primitive), of conductor $q^{*'} > 1$ coprime to Q and dividing some d_i , so the twist condition applies to it. For all other i (and for all i if there is no severe pair), $|\lambda_i - 1| \leq x^{\beta_1-1}/\beta_1 \leq 2E_2$ by the third claim of Lemma 3.2. Since $\lambda_i < 2$ and $x^{\beta^*-1}/\beta^* \leq 1$, splitting $\lambda_i = 1 + (\lambda_i - 1)$ gives

$$\begin{aligned} \sum c_i \theta &\geq \frac{x}{\varphi(Q)} (\mu - 2|\mu_\psi| - M_1(2C_0 E_1 + 2E_2)) \\ &\geq \frac{x}{\varphi(Q)} (\mu/2 - M_1(2C_0 E_1 + 2E_2)), \end{aligned}$$

using $|\mu_\psi| \leq \mu/4$. Both brackets are positive once $E_1, E_2 \leq \mu/(8(C_0 + 1)M_1) = e^{1-K}/(8(C_0 + 1))$. This holds if C_1 is large: $\log x/\log Z \geq C_1 K$, $(\log x)^{3/5}/(\log \log x)^{1/5} \geq (\log x)^{1/2} \geq C_1^{1/2} K$ and $\log x/(13 \log 3Z^2) \geq C_1 K/52$ (as $Z \geq 2$), so each of the three exponentials is at most $e^{-cC_1^{1/2}K}$ with $c > 0$ absolute. Hence $\sum_p \log p \mathbf{1}[W(p) > T] \geq \sum_p \log p B(p) > 0$. $\square$

No lower bound for $1 - \beta^*$ is used: in Case A the exceptional factor λ_* multiplies main term and error alike. The theorem and its proof are taken verbatim from [2].

Remark 4.2 (Assessment). In [2] the Linnik-range uniformity was essential, because there $K \leq T^{o(1)}$ while $\log Z \geq T^{1/k}$. In the present application K and $\log Z$ are both of order $(\log T)^7$ (§7). So a classical prime number theorem of Page type, with error $O(xe^{-c\sqrt{\log x}})$ uniformly for $q \leq e^{c\sqrt{\log x}}$ and with the exceptional zero treated as above, would require $\log x \gg (\log Z + K)^2$. That is the same order $(\log T)^{14}$. We have not written out this variant and do not use it.

5. PRODUCT SYSTEMS AND THE SANDWICH MINORANT

Setting. Let $\mathcal{P}$ be a finite set of N coordinates. Let X_ℓ , $\ell \in \mathcal{P}$, be independent random variables, X_ℓ uniform on a finite set G_ℓ with $|G_\ell| \leq T$; write $\mathbb{P}, \mathbb{E}$ for this product measure. An *event* is a set $E = \{X_\ell = \sigma_\ell \mid \ell \in \text{supp } E\}$ with $\emptyset \neq \text{supp } E \subseteq \mathcal{P}$ and $\sigma \in \prod_{\text{supp } E} G_\ell$. Let $E_1, \dots, E_m$ be events (repetitions allowed) with $|\text{supp } E_i| \leq k$, and put $A_i = \mathbf{1}_{E_i}$, $F_{<i} = \prod_{j < i} (1 - A_j)$, $F = F_{<m+1}$,

$$w_\ell = \sum_{i: \ell \in \text{supp } E_i} \mathbb{P}(E_i), \quad S = \sum_{i \leq m} \mathbb{P}(E_i).$$

By Lemma 2.7, the residual system of Definition 2.6 is an instance with $N \leq T$ and $m \leq T^{k+2}$.

Lemma 5.1 (Local lemma with conditioning; classical). *Let $\mathcal{A}$ be a finite family of events in a product space, each depending on a finite set of coordinates; write $A' \sim A$ if they share a coordinate. Suppose $x_A \in [0, 1)$ satisfy $\mathbb{P}(A) \leq x_A \prod_{A' \sim A, A' \neq A} (1 - x_{A'})$. Then $\mathbb{P}(\bigcap_{A \in \mathcal{A}} \bar{A}) \geq \prod_{A \in \mathcal{A}} (1 - x_A)$. Moreover, for every event B depending on a set J of coordinates, $\mathbb{P}(B \mid \bigcap_{A \in \mathcal{A}} \bar{A}) \leq \mathbb{P}(B) \prod_{A \in \Gamma(B)} (1 - x_A)^{-1}$, where $\Gamma(B)$ is the set of $A \in \mathcal{A}$ depending on some coordinate in J .*

Proof. The first statement is [1, Lemma 5.1.1]. Its proof shows $\mathbb{P}(A \mid \bigcap_{A' \in \mathcal{S}} \bar{A}') \leq x_A$ for every $\mathcal{S} \subseteq \mathcal{A}$ with $A \notin \mathcal{S}$. For the second (cf. [11, Thm. 2.1]), note that B is independent of $\bigcap_{A \notin \Gamma(B)} \bar{A}$, so

$$\mathbb{P}\left(B \mid \bigcap_{A \in \mathcal{A}} \bar{A}\right) \leq \frac{\mathbb{P}(B) \mathbb{P}(\bigcap_{A \notin \Gamma(B)} \bar{A})}{\mathbb{P}(\bigcap_{A \in \mathcal{A}} \bar{A})} = \frac{\mathbb{P}(B)}{\mathbb{P}(\bigcap_{A \in \Gamma(B)} \bar{A} \mid \bigcap_{A \notin \Gamma(B)} \bar{A})}.$$

The chain rule and the displayed claim bound the denominator below by $\prod_{A \in \Gamma(B)} (1 - x_A)$. $\square$

Corollary 5.2 (Proved). *In the Setting, assume $w_\ell \leq 1/(64k)$ for all ℓ . Then $\delta := \mathbb{E}F \geq e^{-2.2S}$. Moreover, for every subfamily $\mathcal{A}$ of $\{E_1, \dots, E_m\}$ and every event B on at most k coordinates, $\mathbb{P}(B \mid \text{no } A \in \mathcal{A} \text{ occurs}) \leq 1.04 \mathbb{P}(B)$.*

Proof. Take $x_{E_i} = 2\mathbb{P}(E_i) \leq 2w_\ell \leq 1/32$ for any $\ell \in \text{supp } E_i$. For an event B on a set J of coordinates, $\sum_{A \in \Gamma(B)} x_A \leq 2 \sum_{\ell \in J} w_\ell \leq |J|/(32k)$. With $B = E_i$ this gives $\prod_{A \sim E_i} (1 - x_A) \geq 31/32$, so the hypothesis of Lemma 5.1 holds for every subfamily. Since $1 - x \geq e^{-1.1x}$ for $0 \leq x \leq 1/32$, we get $\delta \geq \prod_i (1 - 2\mathbb{P}(E_i)) \geq e^{-2.2S}$, and $\prod_{A \in \Gamma(B)} (1 - x_A)^{-1} \leq e^{1.1/32} \leq 1.04$ when $|J| \leq k$. $\square$

Efron–Stein decomposition (classical; [8, 15], [21, §8.3]). For $\varphi : \prod_{\mathcal{P}} G_\ell \rightarrow \mathbb{R}$ and $W \subseteq \mathcal{P}$ let $\mathbb{E}[\varphi \mid X_W]$ be the conditional expectation given the coordinates in W , and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid X_W].$$

Then $\varphi = \sum_U \varphi^{=U}$, the components are mutually orthogonal, and $\varphi^{=U}$ is orthogonal to every function of $X_{U'}$ with $U' \not\supseteq U$. Hence the space of sums of functions of at most t coordinates is $\bigoplus_{|U| \leq t} V_U$, where V_U is the space of all possible $\varphi^{=U}$, and

$$\text{En}(\varphi; t) := \sum_{|U| > t} \|\varphi^{=U}\|_2^2 = \min \left\{ \mathbb{E}(\varphi - g)^2 : g \text{ a sum of functions of } \leq t \text{ coordinates} \right\}, \quad (3)$$

attained at $\varphi^{\leq t} := \sum_{|U| \leq t} \varphi^{=U}$.

For $j \leq m$ let $F^{(j)}$ be the function of the coordinates outside $\text{supp } E_j$ obtained from $F_{<j}$ by fixing $X_\ell = \sigma_{j,\ell}$ for $\ell \in \text{supp } E_j$. Then $A_j F_{<j} = A_j F^{(j)}$, and $F^{(j)}$ is the avoidance indicator of the events E_i ($i < j$) compatible with E_j , restricted to $\mathcal{P} \setminus \text{supp } E_j$. Restricting such an event may leave it with empty support, and then $F^{(j)} \equiv 0$.

Lemma 5.3 (Bazzi–Razborov sandwich [3, 24]; proved). *Let $u_1, \dots, u_m$ be arbitrary real functions, $v_i = \sum_{j < i} A_j u_j$, and*

$$B = 1 - \sum_{i \leq m} A_i (1 - v_i)^2.$$

Then $B \leq F$ pointwise, and with $e_j = F_{<j} - u_j$,

$$F - B = \sum_i A_i \left(\sum_{j < i} A_j e_j \right)^2, \quad \mathbb{E}[F - B] \leq m^2 \sum_{j \leq m} \mathbb{E}[A_j e_j^2].$$

If u_j is a function of the coordinates outside $\text{supp } E_j$, then $\mathbb{E}[A_j e_j^2] = \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$.

Proof. The identity $1 - F = \sum_i A_i F_{<i}$ holds pointwise: the sum counts the first occurring event. Put $f_{<i} = 1 - F_{<i} = \sum_{j < i} A_j F_{<j}$. If $F_{<i} = 1$, then $A_j = 0$ for all $j < i$, so $v_i = 0$ and $A_i(1 - v_i)^2 = A_i = A_i F_{<i}$. If $F_{<i} = 0$, then $f_{<i} = 1$, and $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$. In both cases $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$, and summing over i ,

$$F - B = \sum_i A_i [(1 - v_i)^2 - F_{<i}] = \sum_i A_i (f_{<i} - v_i)^2 \geq 0,$$

with $f_{<i} - v_i = \sum_{j < i} A_j (F_{<j} - u_j)$. By Cauchy–Schwarz, $(\sum_{j < i} A_j e_j)^2 \leq m \sum_{j < i} A_j e_j^2$. Drop $A_i \leq 1$ and sum over i . For the last claim, $A_j e_j^2 = A_j (F^{(j)} - u_j)^2$, and $F^{(j)} - u_j$ is independent of A_j . $\square$

This is Razborov’s simplification [24] of Bazzi’s argument [3]. The choice of u_j as an ℓ^2 -approximation of the “conditional survival function” $F^{(j)}$ is the one used there. We include the short proof for completeness. From now on

$$u_j := (F^{(j)})^{\leq t}, \quad \text{so} \quad \mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \text{En}(F^{(j)}; t). \quad (4)$$

Cells. Now let $G_\ell = (\mathbb{Z}/\ell^{e_\ell})^\times$ with $\ell^{e_\ell} \leq T$, as in Definition 2.6. A *cell* on $I \subseteq \mathcal{P}$ is a set $C = \{X_I = c\}$, $c \in \prod_I G_\ell$. On integers n (with $X_\ell = n \bmod \ell^{e_\ell}$) it is the unit class $n \equiv c \pmod{d_C}$, $d_C = \prod_I \ell^{e_\ell} \leq T^{|I|}$, and $\mathbb{P}(C) = 1/\varphi(d_C)$. For a finite combination $f = \sum_\alpha c_\alpha \mathbf{1}_{C_\alpha}$ put $M_1(f) = \sum_\alpha |c_\alpha| \mathbb{P}(C_\alpha)$; this depends on the representation, which will always be the one obtained by expanding products. If C, D are cells on I, J , then $\mathbf{1}_C \mathbf{1}_D = \mathbf{1}_{C \cap D}$, where $C \cap D$ is empty or a cell on $I \cup J$, both as functions on $\prod G_\ell$ and on $\mathbb{Z}$. Moreover $\mathbb{P}(C \cap D) \leq \mathbb{P}(C) \mathbb{P}(D) \prod_{I \cap J} |G_\ell| \leq \mathbb{P}(C) \mathbb{P}(D) T^{\min(|I|, |J|)}$. Hence

$$M_1(fg) \leq M_1(f) M_1(g) T^{\min(a, b)} \quad (5)$$

if all cells of f (of g) lie on at most a (at most b) coordinates.

Lemma 5.4 (Size of B ; proved). *With u_j as in (4), B is a combination of cells, each on at most $3k + 2t$ coordinates (so each modulus satisfies $d \leq T^{3k+2t}$), and*

$$\log M_1(B) \leq 3 \log m + 4t \log(N+1) + 4(3k+2t)\mathcal{L} + 2.$$

The identity $B = 1 - \sum_i A_i(1 - v_i)^2$ also holds for the expanded cell combination evaluated at any integer n .

Proof. For $\varphi = F^{(j)}$, a function of the $N' \leq N$ coordinates outside $\text{supp } E_j$,

$$\varphi^{\leq t} = \sum_{|U| \leq t} \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid X_W] = \sum_{|W| \leq t} c_W \mathbb{E}[\varphi \mid X_W], \quad c_W = \sum_{i=0}^{t-|W|} (-1)^i \binom{N' - |W|}{i}.$$

Here $|c_W| \leq (N+1)^t$, and there are at most $(N+1)^t$ sets W . Further $\mathbb{E}[\varphi \mid X_W] = \sum_{C \text{ cell on } W} \mathbb{E}[\varphi \mid C] \mathbf{1}_C$ has $M_1 = \mathbb{E}\varphi \leq 1$, as $0 \leq \varphi \leq 1$. So $M_1(u_j) \leq (N+1)^{2t}$, with cells on at most t coordinates. Expanding,

$$B = 1 - \sum_i A_i + 2 \sum_i \sum_{j < i} A_i A_j u_j - \sum_i \sum_{j, j' < i} A_i A_j A_{j'} u_j u_{j'}.$$

Each term is a product of at most five factors: A 's with one cell on $\leq k$ coordinates and $M_1 \leq 1$, and u 's. So every cell lies on at most $3k + 2t$ coordinates. By (5), applied four times with exponent at most $3k + 2t$, a term with two u 's has $M_1 \leq (N+1)^{4t} T^{4(3k+2t)}$; terms with fewer factors obey the same bound. Counting the terms with coefficient 2 twice, there are at most $1 + m + 2m^2 + m^3 \leq 5m^3$ terms (if $m \geq 1$; for $m = 0$, $B = 1$). Since $\log 5 < 2$, the bound follows. The last claim holds because the cell identities hold on $\mathbb{Z}$. $\square$

Lemma 5.5 (Twist; proved). *In the situation of Definition 2.6, assume $w_\ell \leq 1/(64k)$ for all $\ell \in \mathcal{P}$ and $\mathbb{E}[F - B] \leq \delta/100$, where $\delta = \mathbb{E}F$. Write $B = \sum_i c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ as in Lemma 5.4, and let $\mu = \sum_i c_i / \varphi(d_i) = \mathbb{E}B$. Let ψ be a real primitive character of conductor $f > 1$ with $\gcd(f, Q_\Pi) = 1$ and $f \mid d_i$ for some i . Then $\mu \geq 0.99\delta$ and*

$$|\mu_\psi| \leq \mu/4, \quad \mu_\psi = \sum_{i: f \mid d_i} c_i \psi(b_i) / \varphi(d_i).$$

Proof. $\mu = \mathbb{E}F - \mathbb{E}[F - B] \geq 0.99\delta$. As $2 \nmid f$ and ψ is real primitive, f is squarefree. Its prime factors are free, since they divide some d_i . View $\psi(n)$ as a function of the coordinates X_ℓ , $\ell \mid f$. For a cell C with modulus d , $\mathbb{E}[\mathbf{1}_C \psi] = \psi(b)/\varphi(d)$ if $f \mid d$. Otherwise some $\ell \mid f$ has X_ℓ free in C , and the nontrivial ℓ -component of ψ averages to 0. Hence $\mu_\psi = \mathbb{E}[B\psi]$, and

$$|\mu_\psi| \leq \mathbb{E}|B - F| + |\mathbb{E}[F\psi]| = \mathbb{E}[F - B] + |\mathbb{E}[F\psi]|.$$

Fix a prime $\ell_0 \mid f$ and let F' be the avoidance indicator of the events E_i with $\ell_0 \notin \text{supp } E_i$. Given $X_{-\ell_0} = (X_\ell)_{\ell \neq \ell_0}$, we have $F = F' \cdot \mathbf{1}[X_{\ell_0} \notin \text{Forb}]$, where Forb is the set of values σ_{i, ℓ_0} of the events $E_i \ni \ell_0$ whose other conditions hold.

The ℓ_0 -component ψ_0 of ψ is a nontrivial character of X_{ℓ_0} , and $F' \cdot \prod_{\ell \neq \ell_0} \psi_\ell$ depends on $X_{-\ell_0}$ only. So

$$|\mathbb{E}[F\psi]| = \left| \mathbb{E} \left[F' \prod_{\ell \neq \ell_0} \psi_\ell \mathbb{E}_{X_{\ell_0}} [\psi_0 \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \right] \right| \leq \sum_{i: \ell_0 \in \text{supp } E_i} \frac{\mathbb{P}(F'=1, E_i \setminus \ell_0)}{|G_{\ell_0}|},$$

where $E_i \setminus \ell_0$ is the event given by the conditions of E_i off ℓ_0 . By Corollary 5.2 applied to the family defining F' , $\mathbb{P}(E_i \setminus \ell_0 \mid F' = 1) \leq 1.04 \mathbb{P}(E_i \setminus \ell_0)$. Hence $|\mathbb{E}[F\psi]| \leq 1.04 w_{\ell_0} \mathbb{E}F' \leq 0.02 \mathbb{E}F'$. Likewise $\mathbb{E}F' - \mathbb{E}F = \mathbb{E}[F' \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \leq 0.02 \mathbb{E}F'$, so $\mathbb{E}F' \leq \delta/0.98$. Altogether $|\mu_\psi| \leq 0.01\delta + 0.0205\delta \leq 0.031\mu/0.99 < \mu/4$. $\square$

6. THE EFRON–STEIN TAIL VIA THE SWITCHING LEMMA

For $f : \{0, 1\}^n \rightarrow \mathbb{R}$ write $f = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S$ with $\chi_S(x) = (-1)^{\sum_{s \in S} x_s}$, and $W^{\geq d}[f] = \sum_{|S| \geq d} \hat{f}(S)^2$. A p -random restriction $\rho = (J, z)$ chooses each coordinate independently to be free (in J) with probability p , and fixes the others to independent uniform bits z . The restricted function f_ρ is a function of x_J . $\text{DT}(g)$ is the decision-tree depth of g . A decision tree of depth $< s$ computes a function of Fourier degree $< s$.

Theorem 6.1 (Håstad’s switching lemma [12]; cited in the form of [21, §4.4]). *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be computed by a DNF of width at most w , and let ρ be a p -random restriction. Then $\Pr_\rho[\text{DT}(f_\rho) \geq s] \leq (5pw)^s$ for every $s \geq 0$.*

Any absolute constant in place of 5 would do. We also use two classical facts. (1) For fixed J and $S \subseteq J$, the coefficient of $f_{(J,z)}$ at S is $\sum_{U \cap J = S} \hat{f}(U) \chi_{U \setminus J}(z)$. By Parseval in z , therefore, $\mathbb{E}_z[\widehat{f_{(J,z)}}(S)^2] = \sum_{U \cap J = S} \hat{f}(U)^2$. (2) A median of the binomial law $\text{Bin}(n, p)$ is $\lfloor np \rfloor$ or $\lceil np \rceil$ [16], so $\Pr[\text{Bin}(n, p) \geq \lfloor np \rfloor] \geq 1/2$.

Lemma 6.2 (Fourier tail; proved modulo Theorem 6.1). *In the Setting of §5, let φ be the avoidance indicator of a family of events, each on at most k coordinates, on a set of $N' \leq N$ coordinates. Let b be an integer with $2^b \geq 4NT$. Then for every integer $k_0 \geq 1$,*

$$\text{En}(\varphi; 10kbk_0) \leq 4 \cdot 2^{-k_0}.$$

Proof. If some event has empty support then $\varphi \equiv 0$, and there is nothing to prove. Otherwise:

(a) *Encoding.* For each coordinate ℓ enumerate $G_\ell = \{g_0, \dots, g_{q-1}\}$ ($q = q_\ell \leq T$) and put $\pi_\ell(u) = g_{\lfloor q \cdot \text{int}(u)/2^b \rfloor}$ for $u \in \{0, 1\}^b$, where $\text{int}(u) \in [0, 2^b)$ is the integer with binary digits u . Each fibre $\pi_\ell^{-1}(g_a)$ corresponds to an interval of integers. Since $2^b \geq q$, the fibre is nonempty, and it has $\lfloor 2^b/q \rfloor$ or $\lceil 2^b/q \rceil$ elements. An interval in $[0, 2^b)$ is a disjoint union of at most $2b$ dyadic intervals, i.e. of subcubes of $\{0, 1\}^b$ given by fixing at most b bits. Let $U = (U_\ell)$ be uniform on $(\{0, 1\}^b)^{N'}$, put $\pi(U) = (\pi_\ell(U_\ell))_\ell$ and $\tilde{\varphi} = \varphi \circ \pi$. The event $\{X_\ell = \sigma_\ell \mid \ell \in \text{supp } E\}$ pulls back to a conjunction

over $\ell \in \text{supp } E$ of disjunctions of conjunctions of at most b literals in the bits of U_ℓ . Distributing, it is a DNF of width $\leq kb$. Hence $f := 1 - \tilde{\varphi}$, the OR of these over all events, is a DNF of width $w \leq kb$.

(b) *Concentration* (Linial–Mansour–Nisan [17]). Take $p = 1/(10w)$ and $d = k_0/p = 10wk_0$. By Theorem 6.1, $\Pr_\rho[\text{DT}(f_\rho) \geq k_0] \leq 2^{-k_0}$. Since $W^{\geq k_0}[f_\rho] \leq \mathbb{E}f_\rho^2 \leq 1$ always, and $W^{\geq k_0}[f_\rho] = 0$ when $\text{DT}(f_\rho) < k_0$, fact (1) gives

$$2^{-k_0} \geq \mathbb{E}_\rho W^{\geq k_0}[f_\rho] = \sum_U \hat{f}(U)^2 \Pr[|U \cap J| \geq k_0] \geq \frac{1}{2} \sum_{|U| \geq d} \hat{f}(U)^2.$$

In the last step, $|U \cap J| \sim \text{Bin}(|U|, p)$ and $\lfloor p|U| \rfloor \geq k_0$, so fact (2) applies. As $\hat{\varphi}(U) = -\hat{f}(U)$ for $U \neq \emptyset$, the function $\tilde{g} = \sum_{|S| < d} \hat{\varphi}(S) \chi_S$ satisfies $\mathbb{E}(\tilde{\varphi} - \tilde{g})^2 \leq 2^{1-k_0}$.

(c) *Back to residues*. Put $g(x) = \mathbb{E}[\tilde{g}(U) \mid \pi(U) = x]$. The blocks U_ℓ are independent and π acts blockwise. Hence $\mathbb{E}[\chi_S(U) \mid \pi(U) = x]$ is the product over the blocks ℓ met by S of $\mathbb{E}[\chi_{S \cap \text{block } \ell}(U_\ell) \mid \pi_\ell(U_\ell) = x_\ell]$, a function of fewer than d coordinates. So g is a sum of functions of at most $d = 10wk_0 \leq 10kbk_0$ coordinates. Let π_* be the law of $\pi(U)$. As $\tilde{\varphi}$ is a function of $\pi(U)$, Jensen gives $\mathbb{E}_{\pi_*}(\tilde{\varphi} - g)^2 = \mathbb{E}[\mathbb{E}[\tilde{\varphi} - \tilde{g} \mid \pi(U)]^2] \leq 2^{1-k_0}$. Finally, the density of the uniform law relative to π_* is $\prod_\ell 2^b / (q_\ell |\pi_\ell^{-1}(x_\ell)|) \leq \prod_\ell (1 - q_\ell 2^{-b})^{-1} \leq (1 - 1/(4N'))^{-N'} \leq 4/3$. So by (3), $\text{En}(\varphi; d) \leq \mathbb{E}(\varphi - g)^2 \leq \frac{4}{3} \cdot 2^{1-k_0} \leq 4 \cdot 2^{-k_0}$. $\square$

The width kb enters t only linearly. Nothing about the events is used beyond their support size; in particular there is no codegree hypothesis.

Corollary 6.3 (Proved modulo Theorem 6.1). *In the Setting of §5 with $N \leq T$ and $m \geq 1$, assume $w_\ell \leq 1/(64k)$ for all ℓ . Let*

$$b = \lceil \log_2(4T^2) \rceil, \quad k_0 = \lceil 3S \log_2 e + \log_2(400m^2(S+1)) \rceil, \quad t = 10kbk_0,$$

and define B by Lemma 5.3 with u_j as in (4). Then $\mathbb{E}[F - B] \leq \delta/100$.

Proof. Here $\log_2$ is the binary logarithm, and $2^b \geq 4T^2 \geq 4NT$. Apply Lemma 6.2 to each $F^{(j)}$: it is the avoidance indicator of events on at most k coordinates. This gives $\text{En}(F^{(j)}; t) \leq 4 \cdot 2^{-k_0} \leq e^{-3S}/(100m^2(S+1))$. By (4), $\mathbb{E}[F - B] \leq m^2 S e^{-3S}/(100m^2(S+1)) \leq e^{-3S}/100$, and $\delta \geq e^{-2.2S}$ by Corollary 5.2. $\square$

7. PROOF OF THE MAIN THEOREMS

Theorem 7.1 (Proved modulo Theorems 3.1 and 6.1). *Let T be large, $7 \leq z \leq T$, $k = \lfloor \mathcal{L}/\log z \rfloor$, and let Π be the output of Lemma 2.5 with $c_0 = 1/(64k)$. Put*

$$k_0 = \lceil 3S^* \log_2 e + \log_2(400T^{2k+4}(S^* + 1)) \rceil, \quad t = 10k \lceil \log_2(4T^2) \rceil k_0.$$

Then there is a prime $p \equiv 1 \pmod{840}$ with $p > T$, $W(p) > T$ and

$$\log p \leq C_1 K \max(\log Z, K), \quad K \leq 2.2 S^* + C'(k+t)\mathcal{L},$$

$$\log Z \leq (\pi(z) + 64k^2 S^*)\mathcal{L} + 2(3k + 2t + 1)\mathcal{L} + 4,$$

with C_1 from Theorem 4.1 and C' absolute.

Proof. Take the residual system of Definition 2.6 for this Π . By Lemma 2.7, $N \leq T$, $m \leq T^{k+2}$, $S \leq S^*$, every support has at most k primes, and $w_\ell \leq 1/(64k)$. If $m = 0$ put $B = 1$. Otherwise define B by Lemma 5.3 with u_j as in (4) and t as above. Since k_0 is increasing in S and m , it is at least the k_0 of Corollary 6.3, and $\text{En}(\cdot; t)$ decreases in t . So Corollary 6.3 gives $\mathbb{E}[F - B] \leq \delta/100$, and Lemma 5.5 gives $\mu \geq 0.99\delta \geq 0.99e^{-2.2S}$ (Corollary 5.2) and the twist condition for every real primitive ψ with conductor coprime to Q_Π dividing some d_i . By Lemma 5.4, every modulus is $d_i \leq T^{3k+2t}$, and

$$\log(M_1/\mu) \leq 3(k+2)\mathcal{L} + 4t(\mathcal{L} + 1) + 4(3k + 2t)\mathcal{L} + 2 + 2.2S^* + 0.02.$$

Pointwise inequality on integers. For every integer n the values $A_i(n) \in \{0, 1\}$ (with $X_\ell = n \pmod{\ell^{e_\ell}}$) satisfy the hypotheses of Lemma 5.3, and the cell expansion of B is valid on $\mathbb{Z}$ (Lemma 5.4). So $B(n) \leq F(n)$, and by Lemma 2.7(iv), $B(n) \leq \mathbf{1}[W(n) > T]$ for every $n \equiv 1 \pmod{Q_\Pi}$.

Transfer. Let $R = \max(T, \max_i d_i)$ and choose a prime $\ell_{\text{aux}} \in (R, 2R]$. Apply Theorem 4.1 with $Q = Q_\Pi \ell_{\text{aux}}$. Every d_i is composed of free primes, which are $\leq T$ and outside Π , so $\gcd(d_i, Q) = 1$; the b_i are units. The inequality $B \leq \mathbf{1}[W > T]$ holds on $n \equiv 1 \pmod{Q}$. The twist condition holds, as a conductor coprime to Q is coprime to Q_Π . Here $\log Z = \log Q_\Pi + \log \ell_{\text{aux}} + \log \max d_i \leq \log Q_\Pi + \log 2 + 2(3k + 2t)\mathcal{L}$, and $\log Q_\Pi \leq (\pi(z) + 64k^2 S^*)\mathcal{L} + \log 24$ by Lemmas 2.7(i) and 2.5. Theorem 4.1 gives a prime p with $p \equiv 1 \pmod{Q}$, $W(p) > T$ and $\log p \leq C_1 K \max(\log Z, K)$. Since $\ell_{\text{aux}} \mid p - 1$ we have $p > \ell_{\text{aux}} > T$, and $840 \mid Q_\Pi$ gives $p \equiv 1 \pmod{840}$. $\square$

Distinctness of the events plays no role: duplicates are harmless in Lemmas 5.1, 5.3 and 5.5, because the per-prime masses w_ℓ count them.

Proof of Theorem 1.1. Take $z = \mathcal{L}^2$, so $k \leq \mathcal{L}/(2 \log \mathcal{L})$ and $\pi(z) \ll \mathcal{L}^2 / \log \mathcal{L}$. By Lemma 2.4(b), $S^* \ll \mathcal{L}^4 \log \mathcal{L}$. Then $k_0 \ll S^* + k\mathcal{L} \ll \mathcal{L}^4 \log \mathcal{L}$ and $t \ll k\mathcal{L}k_0 \ll \mathcal{L}^6$, so $K \ll \mathcal{L}^7$. Also $\log Q_\Pi \ll (\mathcal{L}^2 / \log \mathcal{L} + k^2 S^*)\mathcal{L} \ll \mathcal{L}^7 / \log \mathcal{L}$ and $\log Z \ll \mathcal{L}^7$. Theorem 7.1 gives a hard prime $p > T$ with $W(p) > T$ and $\log p \leq C\mathcal{L}^{14}$, i.e. $W(p) > T \geq \exp(c(\log p)^{1/14})$. Applying this to a sequence $T_1 < T_2 < \dots$ with T_{j+1} larger than the prime found for T_j gives infinitely many distinct primes. $\square$

Proof of Theorem 1.2. Take $z = \mathcal{L}^2$ again, and use Lemma 2.4(a) instead of (b). Now k_0 , t , K , $\log Q_\Pi$ and $\log Z$ are all at most $\mathcal{L}^{O(1)}(S^* + 1)$, so

$$\log \log p \leq 2 \log(S^* + 1) + O(\log \mathcal{L}) \leq (2 \log 2 + o(1)) \mathcal{L} / \log \mathcal{L}.$$

Since $x \mapsto x/\log x$ is increasing for $x \geq e$, we get $\mathcal{L}/\log \mathcal{L} \geq Y := (1/(2 \log 2) - o(1)) \log \log p$, hence $\mathcal{L} \geq Y \log \mathcal{L} \geq Y \log Y$. That is, $\log W(p) > \mathcal{L} \geq (1/(2 \log 2) - o(1)) \log \log p \cdot \log \log \log p$. Infinitely many primes arise as before. $\square$

Remark 7.2 (Where the exponent comes from; Assessment). The Haar side is $\log(1/\delta) \ll S^* \ll \mathcal{L}^4 \log \mathcal{L}$, plus $\log Q_\Pi \ll \mathcal{L}^7 / \log \mathcal{L}$ for the quarantine; cf. [2], where $\log(1/\delta^*(T)) \ll \mathcal{L}^7 / \log \mathcal{L}$ is proved for the avoider density. The prime side costs $K \cdot \log Z$, with both factors of order $\mathcal{L}^7$: K through $t\mathcal{L}$, and $\log Z$ through both $t\mathcal{L}$ and $\log Q_\Pi$. So the exponent 14 is bookkeeping, not a barrier. Sharper choices of the parameters appear to lower it somewhat; we do not pursue this here. We make no claim about the true order of $W(p)$.

8. SUMMARY OF STATUS

- Proved: Lemmas 2.1–2.7, 5.3, 5.4, 5.5; Corollary 5.2 (with the classical Lemma 5.1); Lemma 2.4(a) (with the classical divisor bound).
- Proved modulo [9, Prop. 1.4]: Lemma 2.4(b).
- Proved modulo Håstad’s switching lemma (Theorem 6.1, a textbook result): Lemma 6.2, Corollary 6.3.
- Proved modulo Thorner–Zaman (Theorem 3.1, with the McCurley statement quoted there): Lemma 3.2, Theorem 4.1, Theorem 7.1 (also using Theorem 6.1), and Theorem 1.2.
- Proved modulo Thorner–Zaman and [9, Prop. 1.4]: Theorem 1.1.
- Assessment: Remarks 4.2 and 7.2.
- Not claimed: anything about the Erdős–Straus conjecture; optimality of the exponent 1/14; any numerical instance.

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